

# Milestone-1

## 1. Theory of Dataset Description

### 1.1 Overview of the Dataset

The dataset used in this project consists of chest X-ray radiographs stored in DICOM format and annotated for thoracic abnormalities. Medical imaging datasets are fundamentally different from traditional image datasets because they contain high-resolution grayscale images along with metadata such as imaging parameters and radiologist annotations. The primary objective of this dataset is to enable automated detection and localization of abnormalities using computer vision techniques.

Chest X-ray images are widely used in healthcare for diagnosing lung diseases and cardiovascular conditions. Each image may contain zero, one, or multiple abnormalities, making the dataset suitable for object detection tasks rather than simple classification problems.

### 1.2 DICOM Image Format

DICOM (Digital Imaging and Communications in Medicine) is the standard file format used in medical imaging systems. Unlike common image formats such as PNG or JPG, DICOM files include:

- Pixel intensity values with high dynamic range
- Metadata describing acquisition details
- Photometric interpretation (MONOCHROME1 or MONOCHROME2)

Because deep learning frameworks and computer vision libraries typically work with standard image formats, preprocessing steps are required to convert DICOM images into PNG images before model training.

### 1.3 Dataset Composition

The dataset includes:

- Thousands of chest X-ray images
- Bounding box annotations created by multiple radiologists
- A metadata file (train.csv) containing object-level information

Each row in the annotation file represents one detected object within an image. Therefore, the number of rows in the dataset is significantly larger than the number of images.

Important attributes in the dataset include:

- `image_id` — unique identifier of each image
- `class_name` and `class_id` — disease category

- `x_min, y_min, x_max, y_max` — bounding box coordinates

Images labeled as “No Finding” contain no visible abnormalities and therefore may not have valid bounding box coordinates.

## **1.4 Dataset Challenges**

Medical imaging datasets introduce several challenges:

- Very high image resolution requiring resizing
- Intensity variations due to acquisition settings
- Multiple annotations per image
- Class imbalance where healthy images dominate

Understanding these characteristics is essential before designing a preprocessing pipeline.

## **2. Theory of Data Preprocessing**

### **2.1 Importance of Preprocessing**

Raw medical images cannot be directly used for deep learning models. Differences in resolution, intensity range, and annotation formats must be standardized to ensure consistent model performance. Preprocessing improves data quality, reduces noise, and prepares images for neural network training.

### **2.2 DICOM to PNG Conversion**

The first preprocessing step involves converting DICOM images into PNG format. This includes:

1. Reading DICOM files using a medical imaging library
2. Extracting pixel arrays
3. Applying min–max normalization
4. Converting pixel values to 8-bit range (0–255)

This conversion improves compatibility with computer vision libraries such as OpenCV and simplifies visualization.

### **2.3 Image Resizing**

Medical images vary greatly in resolution, often exceeding  $2000 \times 2000$  pixels. Training deep learning models on such large images is computationally expensive. Therefore, images are resized to a fixed dimension (e.g.,  $512 \times 512$ ). Resizing ensures:

- Uniform input dimensions
- Reduced memory usage
- Faster processing time

## **2.4 Intensity Normalization**

Medical images contain high dynamic range pixel values. Normalization scales pixel values to a standard range, typically between 0 and 1. This stabilizes gradient updates during training and improves convergence of neural networks.

## **2.5 Contrast Enhancement (CLAHE)**

Chest X-ray images often suffer from low contrast, making subtle abnormalities difficult to detect. Contrast Limited Adaptive Histogram Equalization (CLAHE) enhances local contrast without amplifying noise excessively. This step highlights important anatomical structures while maintaining image quality.

## **2.6 Noise Reduction**

Gaussian blur is applied as a light denoising technique. Medical images may contain acquisition noise, and slight smoothing helps improve feature consistency while preserving essential structures.

## **2.7 Annotation Parsing and Bounding Box Scaling**

Since images are resized during preprocessing, original bounding box coordinates must also be scaled proportionally. Annotation parsing involves:

- Grouping rows by image ID
- Extracting bounding box coordinates
- Scaling coordinates according to the new image size

This ensures that bounding boxes remain correctly aligned with resized images.

## **2.8 Verification of Preprocessing**

To validate the preprocessing pipeline:

- Sample images were processed
- Bounding boxes were extracted and inspected
- Images without abnormalities correctly produced empty bounding box lists

This verification confirms that the preprocessing pipeline is functioning correctly and producing model-ready data.

### **3. Conclusion**

The preprocessing pipeline transforms raw medical imaging data into a structured format suitable for machine learning workflows. Key steps include DICOM conversion, resizing, normalization, contrast enhancement, denoising, and annotation scaling. These processes ensure data consistency and form the foundation for future model training and evaluation stages.